Chapter 4: Continuous Random Variables

Chi-Kin Lam

September 13, 2026

CSAI2121 Probability and Statistics
Macao Polytechnic University

Learning Objectives

By the end of this lecture, you should be able to:

- Explain why continuous random variables require a **density** rather than a mass function
- Construct and interpret a probability density function (PDF)
- Construct and interpret a cumulative distribution function (CDF) for continuous X
- Compute the expected value and variance via integration
- Recognize and apply the Uniform, Exponential, and Normal distributions
- Connect the Exponential distribution to the Poisson process (Chapter 3)

From Discrete to Continuous

Motivation: Why Continuous Random Variables?

- Last week: X took countable values (0, 1, 2, 3, ...).
- Today: X can take **any** real value in an interval.
- Examples: response time of a server, temperature, a model's confidence score $\in [0, 1]$, height, weight.
- These quantities are not naturally "countable" — we need new tools.

Recall: Discrete vs. Continuous

- **Discrete** (Chapter 3): countable values, PMF $p_X(x) = P(X = x)$.
- **Continuous** (this chapter): uncountably infinite values on an interval.

Key Conceptual Shift

For a continuous random variable, $P(X = x) = 0$ for **every** single value x. We can only meaningfully talk about $P(a \leq X \leq b)$.

Why Does $P(X = x) = 0$?

- Intuition: there are infinitely many possible values in any interval, however small.
- If each single point had positive probability, the total over infinitely many points would exceed 1.
- Analogy: a single point has zero *length* on a ruler — but an interval has positive length.

Why Does $P(X = x) = 0$?

- Intuition: there are infinitely many possible values in any interval, however small.
- If each single point had positive probability, the total over infinitely many points would exceed 1.
- Analogy: a single point has zero *length* on a ruler — but an interval has positive length.

Consequence

$$P(a \leq X \leq b) = P(a < X \leq b) = P(a \leq X < b) = P(a < X < b)$$

The endpoints do not matter for continuous X.

Worked Example 1: Response Time (AI Context)

Let $X =$ the response time (in seconds) of an ML inference API to a single request.

Worked Example 1: Response Time (AI Context)

Let $X =$ the response time (in seconds) of an ML inference API to a single request.

- X could be $0.1832...$, $0.1833...$, or any real number ≥ 0.
- Asking "What is $P(X = 0.20000\ldots)$?" is meaningless (answer: 0).
- The useful question is: "What is $P(0.15 \leq X \leq 0.25)$?"

Probability Density Function

Definition of PDF

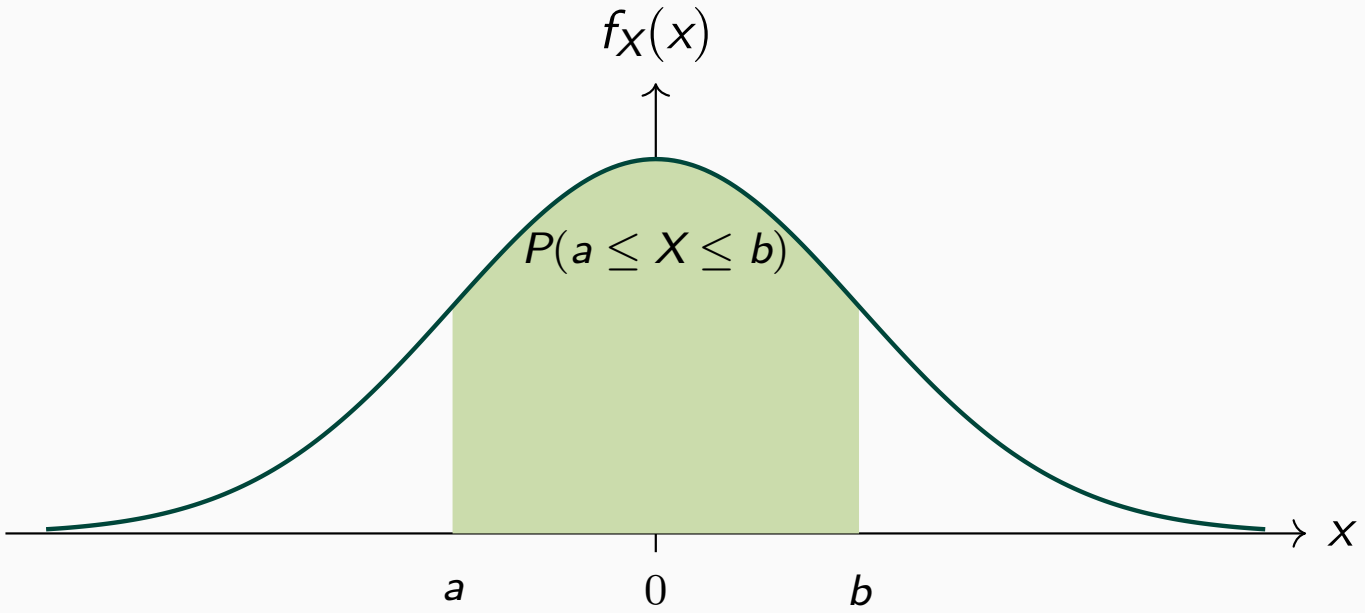
Definition

The **probability density function** (PDF) of a continuous random variable X is a function $f_X(x)$ such that

$$P(a \leq X \leq b) = \int_a^b f_X(x)\,dx$$

- Property 1: $f_X(x) \geq 0$ for all x
- Property 2: $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$
- $f_X(x)$ itself is **not** a probability — it can exceed 1!

Visual Intuition: PDF as Area Under a Curve



- The shaded **area** between a and b equals $P(a \leq X \leq b)$.
- Contrast with last week's bar chart: bars (discrete) vs. a smooth curve and area (continuous).

Worked Example 2: A Simple Triangular-Style PDF

Let $f_X(x) = 2x$ for $0 \leq x \leq 1$ (and 0 otherwise).

Worked Example 2: A Simple Triangular-Style PDF

Let $f_X(x) = 2x$ for $0 \leq x \leq 1$ (and 0 otherwise). **Check it is a valid PDF:**

$$\int_0^1 2x\,dx = \left[x^2\right]_0^1 = 1 \quad \checkmark$$

Worked Example 2: A Simple Triangular-Style PDF

Let $f_X(x) = 2x$ for $0 \leq x \leq 1$ (and 0 otherwise). **Check it is a valid PDF:**

$$\int_0^1 2x\,dx = \left[x^2\right]_0^1 = 1 \quad \checkmark$$

Compute $P(0.25 \leq X \leq 0.75)$:

$$\int_{0.25}^{0.75} 2x\,dx = \left[x^2\right]_{0.25}^{0.75} = 0.5625 - 0.0625 = 0.5$$

Worked Example 3: Model Confidence Score (AI Context)

A classifier outputs a confidence score $X \in [0, 1]$ with PDF $f_X(x) = 3x^2$.

Worked Example 3: Model Confidence Score (AI Context)

A classifier outputs a confidence score $X \in [0, 1]$ with PDF $f_X(x) = 3x^2$. **Check:** $\int_0^1 3x^2\,dx = \left[x^3\right]_0^1 = 1$ ✓

Worked Example 3: Model Confidence Score (AI Context)

A classifier outputs a confidence score $X \in [0, 1]$ with PDF $f_X(x) = 3x^2$. **Check:** $\int_0^1 3x^2\,dx = \left[x^3\right]_0^1 = 1$ ✓

$$P(X \geq 0.9) = \int_{0.9}^{1} 3x^2\,dx = 1 - 0.9^3 = 1 - 0.729 = 0.271$$

Worked Example 3: Model Confidence Score (AI Context)

A classifier outputs a confidence score $X \in [0, 1]$ with PDF $f_X(x) = 3x^2$. **Check:** $\int_0^1 3x^2\,dx = \left[x^3\right]_0^1 = 1$ ✓

$$P(X \geq 0.9) = \int_{0.9}^{1} 3x^2\,dx = 1 - 0.9^3 = 1 - 0.729 = 0.271$$

Interpretation: about 27% of predictions have confidence above 0.9.

Class Exercise (Pair Work)

Problem

Let $f_X(x) = c(4 - x)$ for $0 \leq x \leq 4$.

1. Find the value of c that makes this a valid PDF.
2. Compute $P(1 \leq X \leq 2)$.

Work in pairs (3 minutes), then we discuss as a class.

Cumulative Distribution Function

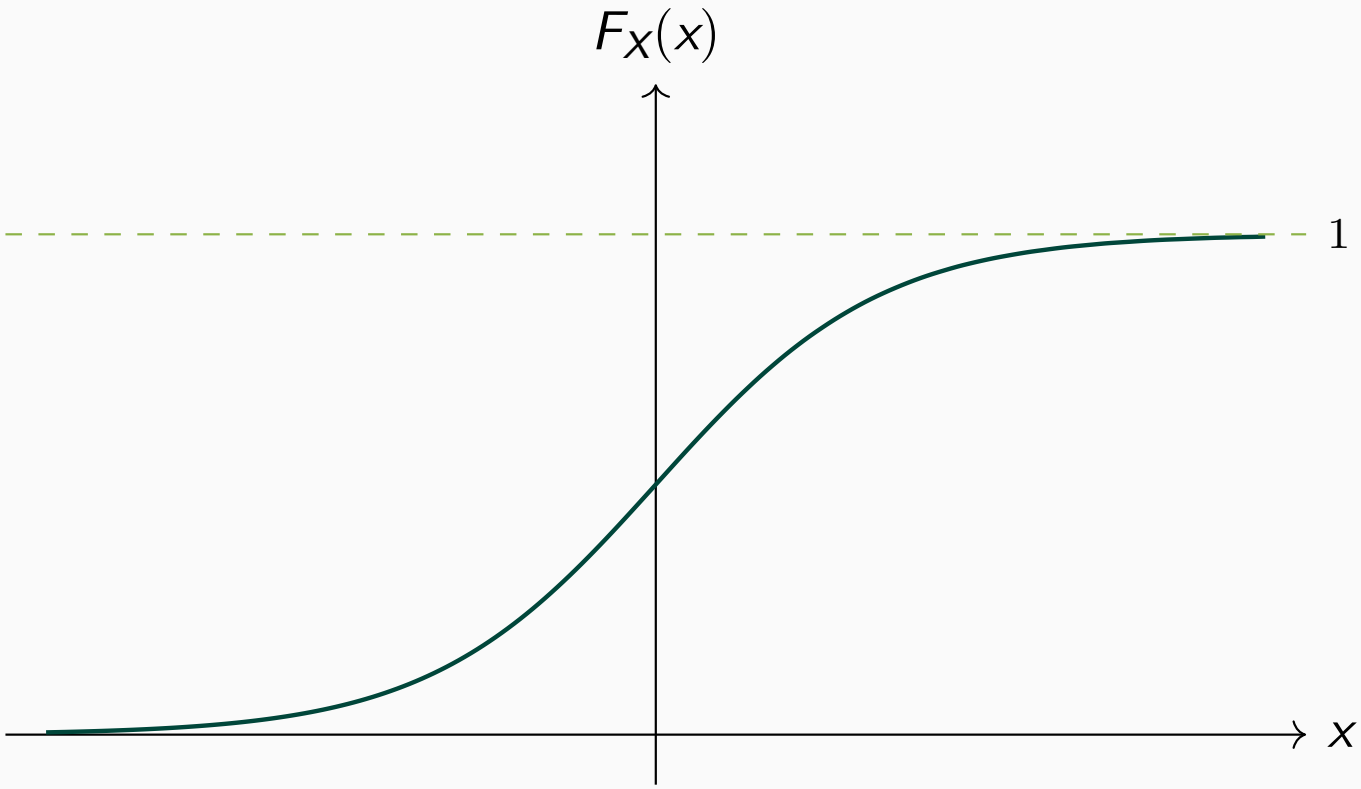
Definition of CDF (Continuous Case)

Definition

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(t)\, dt$$

- F_X is non-decreasing and **continuous** (no jumps, unlike the discrete case!)
- $F_X(-\infty) = 0, \quad F_X(\infty) = 1$
- Fundamental relationship: $f_X(x) = \dfrac{d}{dx} F_X(x)$

Visual Intuition: CDF as a Smooth S-Curve



- Compare with last week's *step* function — continuous X gives a *smooth* curve, no jumps.
- The slope of F_X at any point equals f_X at that point.

Worked Example 4: CDF from a PDF

Recall $f_X(x) = 2x$ for $0 \leq x \leq 1$.

Worked Example 4: CDF from a PDF

Recall $f_X(x) = 2x$ for $0 \leq x \leq 1$.

$$F_X(x) = \int_0^x 2t\,dt = x^2, \qquad 0 \leq x \leq 1$$

Worked Example 4: CDF from a PDF

Recall $f_X(x) = 2x$ for $0 \le x \le 1$.

$$F_X(x) = \int_0^x 2t\,dt = x^2, \qquad 0 \le x \le 1$$

Recompute $P(0.25 \le X \le 0.75)$ using the CDF:

$$F_X(0.75) - F_X(0.25) = 0.75^2 - 0.25^2 = 0.5625 - 0.0625 = 0.5$$

Matches our earlier integration — as expected.

Recovering the PDF from the CDF

Fact

$$f_X(x) = F'_X(x)$$

- Useful when only the CDF is given.
- Example: if $F_X(x) = x^2$ on $[0, 1]$, then $f_X(x) = \frac{d}{dx}x^2 = 2x$ — matches the original PDF.

Class Exercise

Problem

Using the PDF from the earlier exercise, $f_X(x) = c(4 - x)$ on $[0, 4]$ (with c found earlier):

1. Construct the CDF $F_X(x)$.
2. Compute $P(X \leq 2)$.

Expectation and Variance

Definition of Expected Value (Continuous)

Definition

$$E[X] = \int_{-\infty}^{\infty} x \cdot f_X(x)\, dx$$

- Same interpretation as before: a weighted average.
- Sums become integrals — otherwise the concept is identical to Chapter 3.

Worked Example 5: Expectation of the $f(x) = 2x$ Distribution

$$E[X] = \int_0^1 x \cdot 2x \, dx = \int_0^1 2x^2 \, dx = \left[\frac{2x^3}{3}\right]_0^1 = \frac{2}{3}$$

Worked Example 5: Expectation of the $f(x) = 2x$ Distribution

$$E[X] = \int_0^1 x \cdot 2x \, dx = \int_0^1 2x^2 \, dx = \left[\frac{2x^3}{3}\right]_0^1 = \frac{2}{3}$$

Key Point

$E[X] = 2/3$ need not be the "middle" of the interval $[0, 1]$ — it is pulled toward where the density is higher (near $x = 1$ here).

Expectation of a Function of X; Variance

Definitions

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f_X(x)\, dx$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Expectation of a Function of X; Variance

Definitions

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f_X(x)\, dx$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Continuing the example:

$$E[X^2] = \int_0^1 x^2 \cdot 2x\, dx = \left[\frac{x^4}{2}\right]_0^1 = \frac{1}{2}$$

$$\text{Var}(X) = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18} \approx 0.056$$

Linearity Still Holds

Key Properties (unchanged from Chapter 3)

$$E[aX + b] = aE[X] + b, \qquad E[X + Y] = E[X] + E[Y]$$

$$\mathrm{Var}(aX + b) = a^2 \,\mathrm{Var}(X)$$

- These rules do **not** depend on whether X is discrete or continuous.
- This is why we introduced them generally back in Chapter 3.

Instructor Demo 1: Rescaling a Confidence Score

Recall the confidence score X with $f_X(x) = 3x^2$ on $[0, 1]$, with $E[X] = 0.75$ and $\text{Var}(X) = 0.0375$ (you may verify this as extra practice).

Instructor Demo 1: Rescaling a Confidence Score

Recall the confidence score X with $f_X(x) = 3x^2$ on $[0, 1]$, with $E[X] = 0.75$ and $\text{Var}(X) = 0.0375$ (you may verify this as extra practice). Suppose we rescale to a 0–100 scale: $Y = 100X$.

$$E[Y] = 100(0.75) = 75, \qquad \text{Var}(Y) = 100^2(0.0375) = 375$$

Instructor Demo 1: Rescaling a Confidence Score

Recall the confidence score X with $f_X(x) = 3x^2$ on $[0, 1]$, with $E[X] = 0.75$ and $\text{Var}(X) = 0.0375$ (you may verify this as extra practice). Suppose we rescale to a 0–100 scale: $Y = 100X$.

$$E[Y] = 100(0.75) = 75, \qquad \text{Var}(Y) = 100^2(0.0375) = 375$$

Notice variance scales by 100^2, not 100 — the same trap flagged in Chapter 3.

The Uniform Distribution

Motivation: Equally Likely Intervals

- What if every sub-interval of equal length is equally likely?
- Examples: a random number generator on $[0, 1]$; arrival time within a fixed window; random weight initialization in a neural network.

Definition: Uniform Distribution

Definition

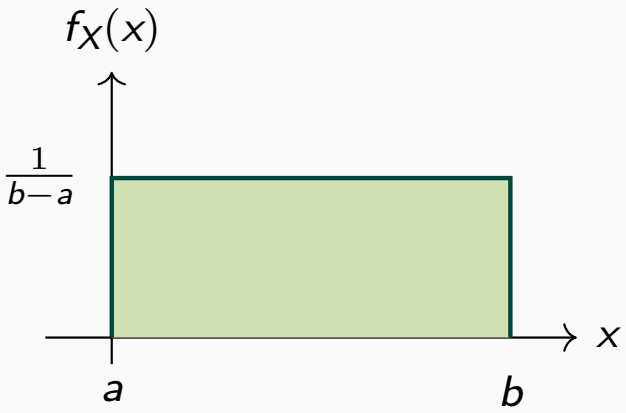
$X \sim \text{Uniform}(a, b)$:

$$f_X(x) = \frac{1}{b-a}, \qquad a \leq x \leq b$$

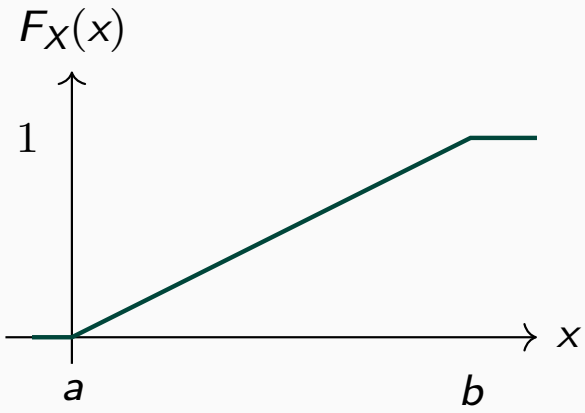
$$E[X] = \frac{a+b}{2}, \qquad \text{Var}(X) = \frac{(b-a)^2}{12}$$

Visual: Uniform PDF and CDF

PDF



CDF



Worked Example 6: Random Number Generator

$X \sim \text{Uniform}(0, 1)$ — the standard output of most random number generators.

Worked Example 6: Random Number Generator

$X \sim \text{Uniform}(0, 1)$ — the standard output of most random number generators.

$$E[X] = \frac{0+1}{2} = 0.5, \qquad \text{Var}(X) = \frac{(1-0)^2}{12} = \frac{1}{12} \approx 0.083$$

Worked Example 6: Random Number Generator

$X \sim \text{Uniform}(0, 1)$ — the standard output of most random number generators.

$$E[X] = \frac{0+1}{2} = 0.5, \qquad \text{Var}(X) = \frac{(1-0)^2}{12} = \frac{1}{12} \approx 0.083$$

$$P(0.2 \leq X \leq 0.7) = 0.7 - 0.2 = 0.5$$

(area = width, since height is constant.)

Worked Example 7: Weight Initialization (AI Context)

A neural network's weights are initialized as $X \sim \text{Uniform}(-0.05, 0.05)$.

Worked Example 7: Weight Initialization (AI Context)

A neural network's weights are initialized as $X \sim \text{Uniform}(-0.05, 0.05)$.

$$E[X] = 0, \qquad \text{Var}(X) = \frac{(0.1)^2}{12} \approx 0.00083$$

Worked Example 7: Weight Initialization (AI Context)

A neural network's weights are initialized as $X \sim \text{Uniform}(-0.05, 0.05)$.

$$E[X] = 0, \qquad \text{Var}(X) = \frac{(0.1)^2}{12} \approx 0.00083$$

Interpretation: initial weights are centered at 0 with small spread — exactly the design goal of common initialization schemes.

Class Exercise

Problem

A queuing system's waiting time is modeled as $X \sim \text{Uniform}(0, 10)$ (minutes).

1. Compute $E[X]$ and $\text{Var}(X)$.
2. Compute $P(X > 8)$.

The Exponential Distribution

Motivation: Modeling Waiting Times

- Recall from Chapter 3: the **Poisson** distribution counts *how many* rare events occur in an interval.
- Today's question: *how long do we wait* until the next event?
- This is the natural continuous counterpart to the Poisson process.

Definition: Exponential Distribution

Definition

$X \sim \text{Exponential}(\lambda)$:

$$f_X(x) = \lambda e^{-\lambda x}, \qquad x \geq 0$$

$$F_X(x) = 1 - e^{-\lambda x}, \qquad x \geq 0$$

- $\lambda =$ the same rate parameter as in the Poisson distribution.

Mean and Variance of Exponential

Key Result

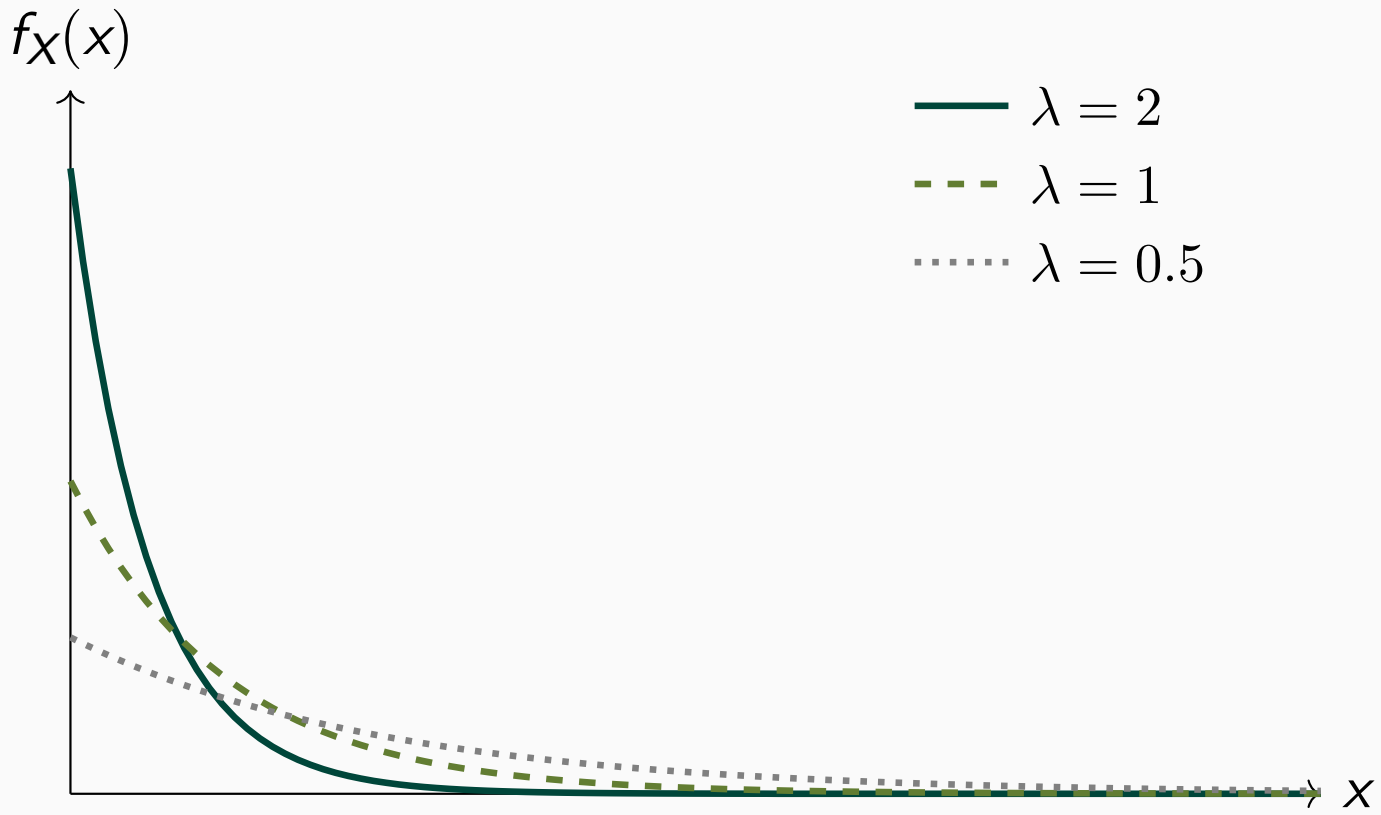
$$E[X] = \frac{1}{\lambda}, \qquad \text{Var}(X) = \frac{1}{\lambda^2}$$

Memoryless Property

$$P(X > s + t \mid X > s) = P(X > t)$$

"Waiting time so far" gives no information about "time remaining" — a distinctive and often surprising property.

Visual: Exponential PDF for Different λ



- Larger $\lambda \Rightarrow$ events happen sooner $\Rightarrow$ steeper decay, shorter average wait.

Worked Example 8: Server Request Waiting Time

Recall Chapter 3, Worked Example 13: server requests arrive at rate $\lambda = 4$ per minute (Poisson). The **time between** requests $T \sim \text{Exponential}(4)$.

Worked Example 8: Server Request Waiting Time

Recall Chapter 3, Worked Example 13: server requests arrive at rate $\lambda = 4$ per minute (Poisson). The **time between** requests $T \sim \text{Exponential}(4)$.

$$E[T] = \frac{1}{4} = 0.25 \text{ min} = 15 \text{ seconds}$$

Worked Example 8: Server Request Waiting Time

Recall Chapter 3, Worked Example 13: server requests arrive at rate $\lambda = 4$ per minute (Poisson). The **time between** requests $T \sim \text{Exponential}(4)$.

$$E[T] = \frac{1}{4} = 0.25 \text{ min} = 15 \text{ seconds}$$

$$P(T > 0.5) = e^{-4(0.5)} = e^{-2} \approx 0.135$$

Worked Example 8: Server Request Waiting Time

Recall Chapter 3, Worked Example 13: server requests arrive at rate $\lambda = 4$ per minute (Poisson). The **time between** requests $T \sim \text{Exponential}(4)$.

$$E[T] = \frac{1}{4} = 0.25 \text{ min} = 15 \text{ seconds}$$

$$P(T > 0.5) = e^{-4(0.5)} = e^{-2} \approx 0.135$$

Interpretation: about 13.5% chance we wait more than 30 seconds for the next request.

Worked Example 9: Time to Next API Failure (AI Context)

Recall Chapter 3, Worked Example 14: an ML inference API fails at rate $\lambda = 1.5$ per hour (Poisson). Let $T =$ time until the next failure.

Worked Example 9: Time to Next API Failure (AI Context)

Recall Chapter 3, Worked Example 14: an ML inference API fails at rate $\lambda = 1.5$ per hour (Poisson). Let $T =$ time until the next failure.

$$T \sim \text{Exponential}(1.5), \qquad E[T] = \frac{1}{1.5} \approx 0.667 \text{ hours} \approx 40 \text{ min}$$

Worked Example 9: Time to Next API Failure (AI Context)

Recall Chapter 3, Worked Example 14: an ML inference API fails at rate $\lambda = 1.5$ per hour (Poisson). Let $T =$ time until the next failure.

$$T \sim \text{Exponential}(1.5), \qquad E[T] = \frac{1}{1.5} \approx 0.667 \text{ hours} \approx 40 \text{ min}$$

$$P(T < 1) = 1 - e^{-1.5} \approx 1 - 0.223 = 0.777$$

Worked Example 9: Time to Next API Failure (AI Context)

Recall Chapter 3, Worked Example 14: an ML inference API fails at rate $\lambda = 1.5$ per hour (Poisson). Let $T =$ time until the next failure.

$$T \sim \text{Exponential}(1.5), \qquad E[T] = \frac{1}{1.5} \approx 0.667 \text{ hours} \approx 40 \text{ min}$$

$$P(T < 1) = 1 - e^{-1.5} \approx 1 - 0.223 = 0.777$$

Connecting the Dots

Same λ, two views: *Poisson* counts failures per hour; *Exponential* describes the waiting time until the next one.

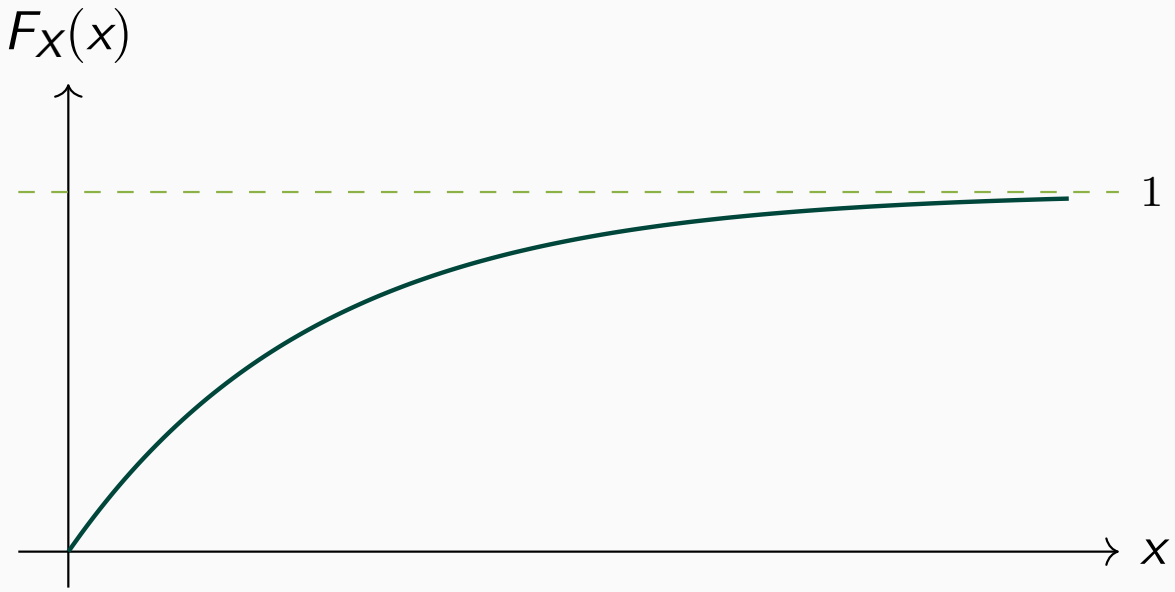
Class Exercise

Problem

A content-moderation system (Chapter 3, Class Exercise) flags posts at rate $\lambda = 3$ per hour. Let $T =$ time until the next flagged post.

1. Compute $E[T]$.
2. Compute $P(T > 0.5)$ (i.e., more than 30 minutes).

Instructor Demo 2: Visualizing the Exponential CDF



- $F_X(x) = 1 - e^{-\lambda x}$ rises quickly at first, then flattens — most of the probability mass is concentrated at small x.
- As $x \to \infty$, $F_X(x) \to 1$, consistent with "eventually, the event will occur."

The Normal (Gaussian) Distribution

Motivation: The Most Important Continuous Distribution

- Appears everywhere: measurement error, exam scores, heights, and — crucially for AI — **residual errors** in regression models.
- Many natural and engineered processes tend toward a bell-shaped distribution (a preview of the Central Limit Theorem, later in the course).

Definition: Normal Distribution

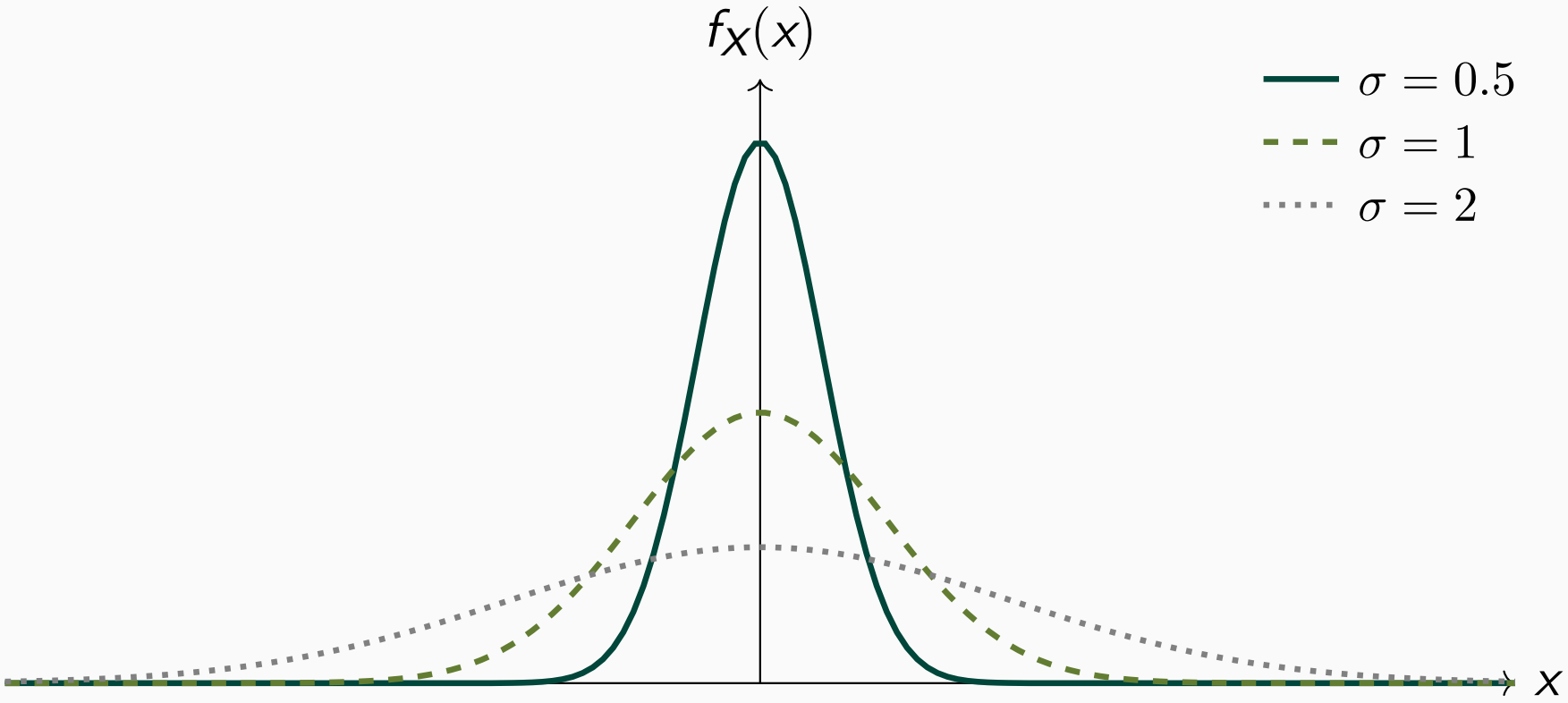
Definition

$X \sim N(\mu, \sigma^2)$:

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad x \in \mathbb{R}$$

$$E[X] = \mu, \qquad \mathrm{Var}(X) = \sigma^2$$

Visual: Normal PDF for Different σ



- All curves are centered at $\mu = 0$; smaller $\sigma \Rightarrow$ taller, narrower peak.

The Standard Normal Distribution

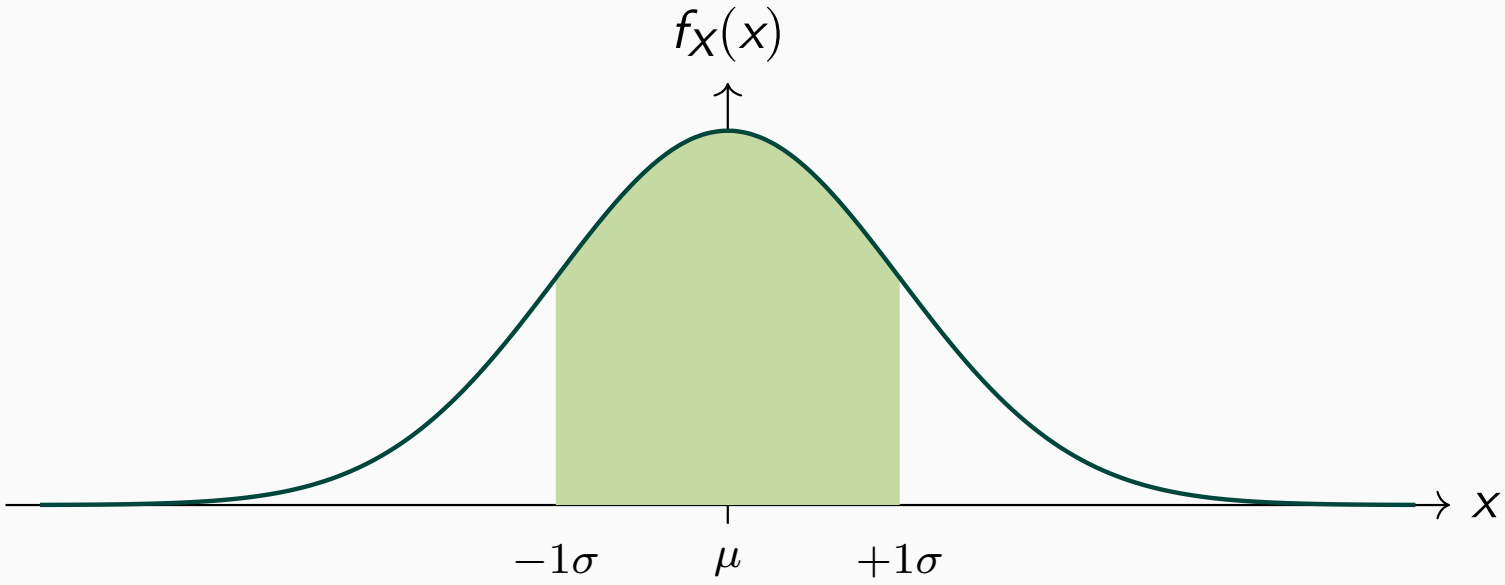
Definition

$Z \sim N(0, 1)$ is the **standard normal**. Any $X \sim N(\mu, \sigma^2)$ can be standardized:

$$Z = \frac{X - \mu}{\sigma}$$

- Standard normal probabilities are looked up in a Z-table (provided in the appendix / online).
- This lets us answer probability questions for *any* Normal distribution using one table.

The Empirical Rule (68–95–99.7)



$$P(\mu - \sigma < X < \mu + \sigma) \approx 68\%,$$

$$P(\mu - 2\sigma < X < \mu + 2\sigma) \approx 95\%,$$

$$P(\mu - 3\sigma < X < \mu + 3\sigma) \approx 99.7\%$$

Worked Example 10: Exam Scores

Exam scores $X \sim N(70, 10^2)$. Find $P(X > 80)$.

Worked Example 10: Exam Scores

Exam scores $X \sim N(70, 10^2)$. Find $P(X > 80)$.

$$Z = \frac{80 - 70}{10} = 1$$

Worked Example 10: Exam Scores

Exam scores $X \sim N(70, 10^2)$. Find $P(X > 80)$.

$$Z = \frac{80 - 70}{10} = 1$$

$$P(X > 80) = P(Z > 1) = 1 - \Phi(1) = 1 - 0.8413 = 0.1587$$

Worked Example 10: Exam Scores

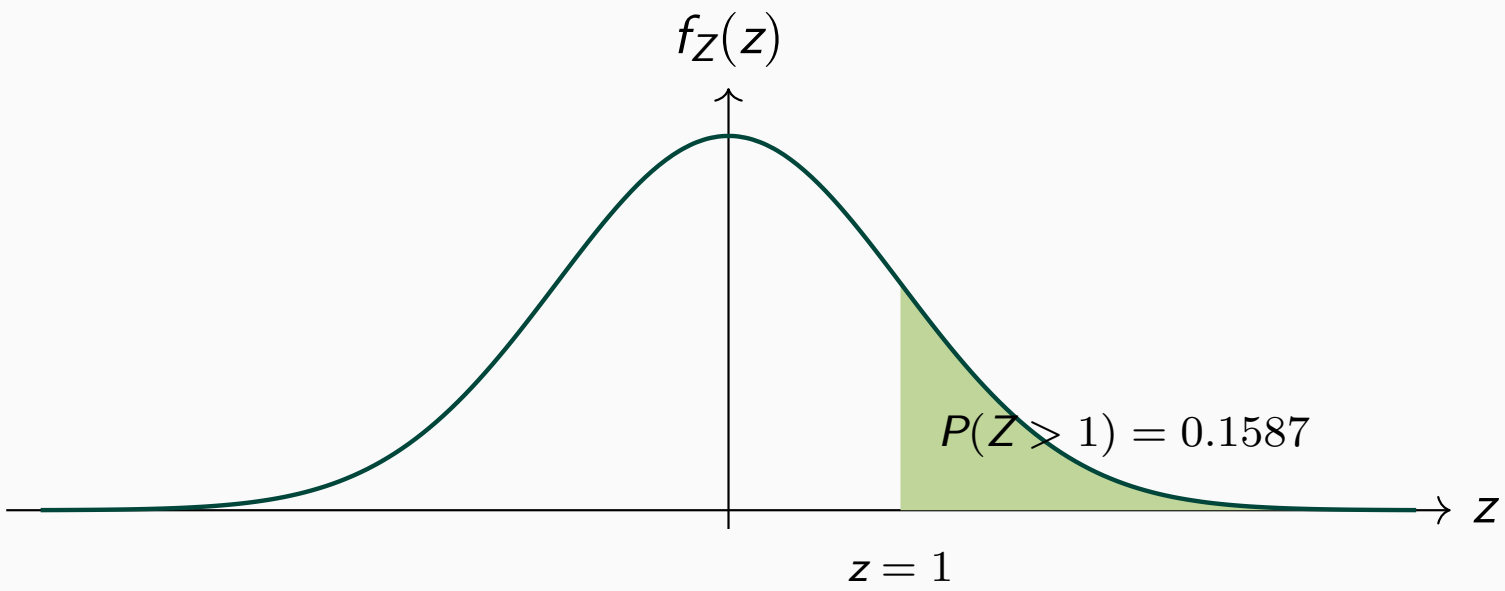
Exam scores $X \sim N(70, 10^2)$. Find $P(X > 80)$.

$$Z = \frac{80 - 70}{10} = 1$$

$$P(X > 80) = P(Z > 1) = 1 - \Phi(1) = 1 - 0.8413 = 0.1587$$

About 15.9% of students score above 80.

Visual: Standardizing and Shading a Tail



Worked Example 11: Regression Residuals (AI Context)

A regression model's prediction errors are assumed $E \sim N(0, 2^2)$ (i.e., $\sigma = 2$).

Worked Example 11: Regression Residuals (AI Context)

A regression model's prediction errors are assumed $E \sim N(0, 2^2)$ (i.e., $\sigma = 2$). **Question:** what is $P(|E| > 3)$ (a "large error")?

$$Z = \frac{3 - 0}{2} = 1.5, \qquad P(Z > 1.5) = 1 - 0.9332 = 0.0668$$

Worked Example 11: Regression Residuals (AI Context)

A regression model's prediction errors are assumed $E \sim N(0, 2^2)$ (i.e., $\sigma = 2$). **Question:** what is $P(|E| > 3)$ (a "large error")?

$$Z = \frac{3 - 0}{2} = 1.5, \qquad P(Z > 1.5) = 1 - 0.9332 = 0.0668$$

$$P(|E| > 3) = 2 \times P(Z > 1.5) = 2(0.0668) = 0.1336$$

Worked Example 11: Regression Residuals (AI Context)

A regression model's prediction errors are assumed $E \sim N(0, 2^2)$ (i.e., $\sigma = 2$). **Question:** what is $P(|E| > 3)$ (a "large error")?

$$Z = \frac{3 - 0}{2} = 1.5, \qquad P(Z > 1.5) = 1 - 0.9332 = 0.0668$$

$$P(|E| > 3) = 2 \times P(Z > 1.5) = 2(0.0668) = 0.1336$$

About 13.4% of predictions have an error exceeding 3 units in either direction — useful for setting alerting thresholds in production ML systems.

Class Exercise

Problem

A sensor's readings are $X \sim N(50, 5^2)$.

1. Compute $P(X < 45)$.
2. Compute $P(45 < X < 60)$.

Work individually (5 minutes); use the Z-table.

Instructor Demo 3: From Binomial to Normal (Preview)

Recall Chapter 3: a Binomial(n, p) with large n and moderate p produces a symmetric, bell-shaped PMF (Slide 37, $n = 10, p = 0.5$).

Instructor Demo 3: From Binomial to Normal (Preview)

Recall Chapter 3: a Binomial(n, p) with large n and moderate p produces a symmetric, bell-shaped PMF (Slide 37, $n = 10, p = 0.5$).

Looking Ahead

This resemblance is not a coincidence — it is an early glimpse of the **Central Limit Theorem**, which formally explains why sums of many random variables tend toward a Normal shape. We will return to this later in the course.

Summary and Looking Ahead

Summary Table: Three Key Continuous Distributions

Distribution	PDF	Mean	Variance	Typical Use
Uniform(a, b)	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	equally likely values in a range
Exponential(λ)	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	waiting time to next rare event
Normal(μ, σ^2)	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2	measurement error, residuals, natural variation

Common Mistakes

- Treating $f_X(x)$ itself as a probability (it is a *density*; only areas are probabilities)
- Forgetting that $P(X = x) = 0$ for continuous X — so $P(X \leq x) = P(X < x)$
- Confusing the Exponential's λ (a *rate*) with the Normal's σ (a *spread*)
- Forgetting to standardize ($Z = (X - \mu)/\sigma$) before using the Z-table

Summary: Key Formulas from Today

$$\int_{-\infty}^{\infty} f_X(x)\,dx = 1, \qquad P(a \le X \le b) = \int_a^b f_X(x)\,dx$$

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt, \qquad f_X(x) = F_X'(x)$$

$$E[X] = \int_{-\infty}^{\infty} x\, f_X(x)\,dx, \qquad \mathrm{Var}(X) = E[X^2] - (E[X])^2$$

$$Z = \frac{X - \mu}{\sigma} \sim N(0,1)$$

Connections Ahead

- Poisson (Chapter 3) $\leftrightarrow$ Exponential (this week): counting events vs. waiting for them
- Binomial's bell shape (Chapter 3) $\rightarrow$ hints at the Central Limit Theorem
- Expectation & Variance $\rightarrow$ still the core machinery for MLE and Confidence Intervals later in the course
- Normal distribution $\rightarrow$ foundational for almost all inferential statistics that follow

Preview of Week 5: Joint Distributions

- Today: a single random variable X.
- Next week: **two or more** random variables considered together — $P(X = x, Y = y)$.
- We will meet joint PMFs/PDFs, marginal distributions, and (in)dependence between random variables — essential groundwork for multivariate ML models.